HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Python**-based agentic workflows for internal quality assurance systems, integrating LLM tool use for automated test case generation and analysis.
- Developed **Python** backend services using **FastAPI** to serve ML models, managing model inference requests and structured output parsing for downstream applications.
- Engineered feedback loops that capture model failures and translate them into actionable insights for prompt refinement and data augmentation, improving system accuracy by approximately **15%** over a 3-month period.
- Collaborated with international quality engineers and translators to gather requirements for AI-powered localization testing tools, translating business needs into technical specifications for LLM integration.
- Fine-tuned a **pre-trained transformer model** on a dataset of ~1.2 million localized software strings to improve contextual understanding and reduce translation errors, achieving a **12%** reduction in human review cycles.
- Built and maintained **CI/CD pipelines** using **Jenkins** and **Git** for the automated deployment of LLM-based applications to staging and production environments.
- Containerized ML inference services using **Docker** and deployed them to a managed Kubernetes cluster, configuring resource limits and auto-scaling policies to handle variable loads.

SKILLS

ML Engineering – LLMs, Prompt Engineering, Model Evaluation, Fine-tuning, Agentic Design, Structured Output, Python, FastAPI, Scikit-learn, Transformer Models

Programming Languages – Python, Swift, Java

Backend Development – FastAPI, Spring Boot, REST APIs, Microservices

ML Operations – CI/CD, Jenkins, Docker, Kubernetes, Model Serving, Feedback Loops

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Data & Analytics – SQL, Pandas, NumPy, Data Modelling, Data Warehousing

Tools & Methodologies – Git, Jira, Confluence, Agile, Scrum

PROJECTS

LLM-Powered Test Case Generation for Software Localization

- Developed a **Python** application that interfaces with an LLM to generate diverse test cases based on software localization requirements, covering over **20** different languages.
- Implemented prompt engineering strategies to guide the LLM in producing structured test scenarios, including expected outputs and edge cases, improving test coverage by an estimated **18%**.
- Integrated the generated test cases into an existing automated testing framework, enabling faster iteration cycles for international software releases.

Automated Screenshot Analysis for UI Consistency

- Engineered a **Python** pipeline that extracts UI elements from screenshots using **computer vision** techniques and compares them against reference images for consistency across different language versions.
- Trained a **convolutional neural network** on a dataset of ~50,000 UI component images to classify element types and detect visual discrepancies, achieving **0.93** accuracy.
- Built a **FastAPI** service to expose the analysis results, providing detailed reports on UI deviations for localization and QA teams.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)